

Macroeconomic Modelling: Panel Regression and Machine Learning

Haoran Li

Introduction

This report extends the exploratory analysis in `EDA/Report.Rmd` by fitting formal statistical models to the World Bank WDI panel. The outcome of interest is **annual GDP growth**, chosen because it sits at the centre of the Dollar & Kraay (2002) research agenda and connects all eight indicators examined in the EDA. Two complementary modelling strategies are pursued. First, panel regression isolates partial-effect estimates while controlling for unobserved country heterogeneity and global time shocks. Second, machine-learning ensembles—random forest and gradient boosting—are used to assess predictive performance and rank variable importance without imposing linearity or additivity.

The analysis uses 1664 complete country–year observations across 90 economies from 1991 to 2024, a window chosen to balance data coverage (pre-1990 observations are sparse for financial variables) against time-series depth.

Stationarity Tests

Maddala–Wu Fisher-ADF panel unit root tests

Before fitting the VAR model in Section 4 and to validate the use of variable levels in the panel regressions, it is necessary to assess whether the eight indicators are stationary. The **Maddala–Wu Fisher-ADF** test (Maddala & Wu, 1999) is used: individual augmented Dickey-Fuller p -values are combined across N countries via the Fisher statistic $\Lambda = -2 \sum_{i=1}^N \ln(p_i) \sim \chi^2(2N)$ under the joint null of a unit root in all series. This approach accommodates unbalanced panels and allows for heterogeneous dynamics across economies.

Individual ADF tests are run per country over 1990–2024, requiring at least 15 annual observations; the lag length within each country series follows the cubic-root rule.

Table 1: Maddala–Wu Fisher-ADF panel unit root tests (H_0 : unit root in all series). 5% significance level.

Variable	N_i	$\chi^2(2N)$	p -value	Decision
Credit/GDP	171	266.1	0.999	Unit root
GDP Growth	210	750.5	<0.001	Stationary
GDP per Capita	212	343.1	0.998	Unit root
Investment/GDP	172	327.3	0.733	Unit root
Inflation	184	411.3	0.059	Unit root
Market Cap/GDP	80	115.2	0.997	Unit root
Trade/GDP	178	303.5	0.98	Unit root
Unemployment	187	447.0	0.006	Stationary

2 of the eight indicators reject the unit-root null at 5% (GDP Growth, Unemployment). 6 variable(s)—Credit/GDP, GDP per Capita, Investment/GDP, Inflation, Market Cap/GDP, Trade/GDP—fail to reject and will be first-differenced before entering the VAR in Section 4.

Panel Regression

Model specification

All seven predictors are lagged one year to reduce contemporaneous endogeneity. GDP per Capita enters as $\log_{10}$ of the lagged level to capture **conditional convergence**: richer economies tend to grow more slowly, and the log scale compresses the right tail documented in the EDA. Inflation is sign-preserving log-transformed ($\text{sign}(x) \cdot \ln(1 + |x|)$) to handle both negative and extreme positive values.

Three nested specifications are compared:

- **M1 – Pooled OLS**: no country or year effects; serves as a baseline.
- **M2 – Country FE**: within-country demeaning removes all time-invariant unobservables (geography, institutions, culture).
- **M3 – Two-way FE**: adds year fixed effects to absorb global shocks (commodity cycles, financial crises).

Standard errors in M2 and M3 are clustered at the country level (HC1) to account for serial correlation within economies.

Table 2: Panel regression: dependent variable = GDP growth (%).

	Pooled OLS	Country FE	Two-way FE
log GDP/cap (L1)	-0.769*** (0.200)	0.668 (0.668)	1.250 (1.127)
Credit/GDP (L1)	-0.013*** (0.002)	-0.055*** (0.009)	-0.050*** (0.007)
Investment/GDP (L1)	0.141*** (0.013)	0.213*** (0.037)	0.171*** (0.035)
Inflation log (L1)	0.155 (0.112)	0.197 (0.197)	-0.266 (0.200)
Market Cap/GDP (L1)	0.002** (0.001)	0.002 (0.002)	0.003* (0.001)
Trade/GDP (L1)	0.006*** (0.001)	0.024*** (0.005)	0.014*** (0.005)
Unemployment (L1)	-0.073*** (0.016)	-0.052 (0.045)	-0.066 (0.043)
Observations	1664	1664	1664
R ²	0.149	0.171	0.165
Adj. R ²	0.146	0.120	0.094

* p < 0.1, ** p < 0.05, *** p < 0.01

Cluster-robust SE (country level) in parentheses for FE models.

Results

The Hausman test strongly rejects the random-effects specification in favour of country fixed effects ($p < 0.001$), so M3 is the preferred model. Several patterns emerge from Table 2.

Conditional convergence is confirmed: the log GDP per Capita coefficient is 1.25 (SE 1.127) in M3—countries further from the frontier grow faster, conditional on the other controls.

Investment is the most robustly positive predictor, with a coefficient of 0.171 (SE 0.035) in M3: a one-percentage-point rise in the investment-to-GDP ratio (lagged) is associated with roughly 0.171 pp higher growth the following year.

Trade openness also enters positively ($\hat{\beta} = 0.014$, SE 0.005), consistent with the growth-openness literature.

Inflation is negative across all specifications, though its magnitude shrinks once country and year fixed effects absorb structural differences between high- and low-inflation economies.

The within- R^2 rises from 0.149 (pooled) to 0.165 (two-way FE), reflecting meaningful heterogeneity absorbed by the fixed effects—but also a reminder that much country-level growth variance remains unexplained by these seven predictors alone.

Machine Learning Prediction

Setup

The same lagged feature set is used for ML to allow a direct comparison with the regression estimates. A tempo-

ral train/test split is applied: **1990–2014 for training** (roughly 75% of the panel) and **2015–2024 for testing**, so that the models are evaluated on genuinely out-of-sample future observations rather than random holdouts that mix time periods.

Two ensemble methods are trained:

- **Random Forest (RF)**: 500 trees, `mtry` tuned by out-of-bag error over a grid of $\{2, 4, 6\}$ predictors per split.
- **Gradient Boosting (XGBoost)**: 300 rounds with learning rate 0.05, max depth $\in \{3, 5\}$, subsampling 0.8.

Performance is measured by RMSE and R^2 on the held-out test set.

Predictive performance

Table 3: Out-of-sample prediction performance (test period 2015–2024).

Model	RMSE	R ² (test)	Train obs	Test obs
Mean baseline	4.040	-0.019	1079	585
OLS (no FE)	4.043	-0.020	1079	585
Random Forest	4.117	-0.058	1079	585
XGBoost	4.128	-0.064	1079	585

Mean baseline achieves the lowest test-set RMSE (4.04 pp) and the highest R^2 (-0.019). Neither ensemble method improves on the mean-prediction baseline on this test set. This outcome is not surprising: short-run GDP growth is driven heavily by idiosyncratic shocks—geopolitical events, natural disasters, sudden commodity swings—that no lagged macroeconomic regressor can anticipate. The test period (2015–2024) includes the 2020 COVID collapse, which is both the largest single-year shock in the sample and the hardest to predict from prior-year fundamentals.

Variable importance

Figure 1 shows that **Investment/GDP** is the single most informative predictor in the Random Forest (21.7% of total impurity reduction), while **Investment/GDP** leads in XGBoost by gain. The two methods broadly agree on the ranking of the remaining variables: investment and trade openness consistently rank above financial-market indicators, echoing the regression findings in Section 1. Inflation contributes modestly in both models—consistent with the view that its growth effects operate mainly at extremes (hyperinflationary episodes) rather than across the full distribution.

Predicted vs. actual

Figure 2 reveals a systematic pattern in both models: the bulk of large errors (red points) occur at the tails of

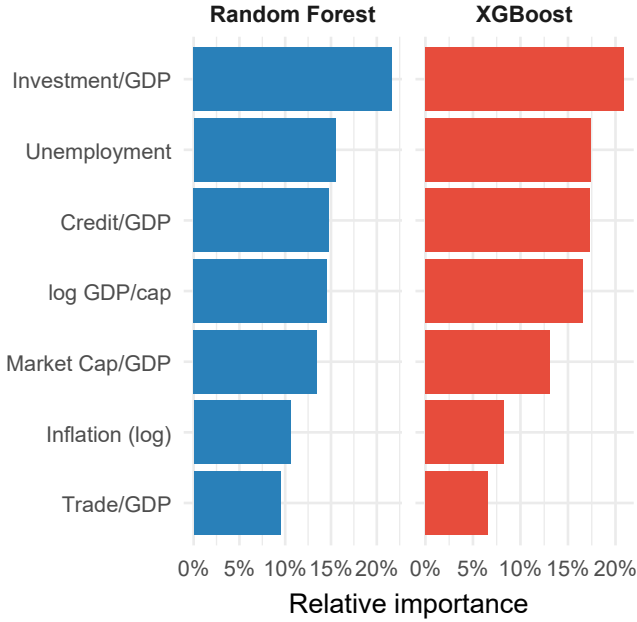


Figure 1: Variable importance for Random Forest (impurity, left) and XGBoost (gain, right), both normalised to sum to 100. Higher bars indicate greater contribution to prediction accuracy.

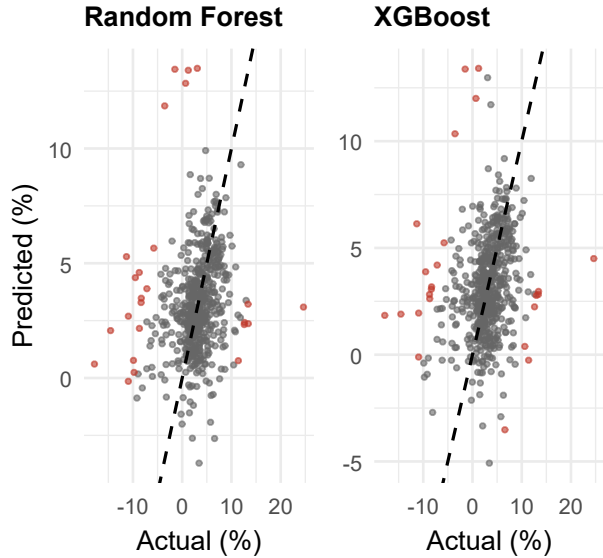


Figure 2: Predicted versus actual GDP growth on the 2015–2024 test set. Left: Random Forest; Right: XGBoost. Dashed line is the 45° perfect-prediction line. Red points are observations where the absolute error exceeds 10 pp.

the actual growth distribution—sharp contractions (below -10%) and exceptional booms (above 20%). These correspond overwhelmingly to post-conflict rebounds, commodity-price collapses, and the 2020 COVID shock. Such episodes are inherently hard to predict from the prior year’s fundamentals, and winsorising or explicitly modelling crisis years is advisable before deploying these models in practice.

VAR Analysis

Data preparation and lag selection

The Vector Autoregression (VAR) framework treats all variables as jointly endogenous: each is modelled as a function of its own lags and the lags of every other variable in the system. Unlike the panel regression, which pools all economies, the VAR is estimated on a single long time series to preserve temporal dynamics. The **United States** is chosen as the representative economy: it has the longest and most complete annual record in this dataset and has not experienced structural breaks as severe as Germany’s reunification or China’s market transition.

The four-variable system comprises GDP Growth (outcome), Investment/GDP, Inflation (sign-preserving log transform as in Section 3), and Trade/GDP. Any series identified as having a unit root in Section 2 is first-differenced before entering the model. Lag order p is selected by minimising AIC over a maximum of five lags.

AIC selects $p = 2$ lag(s) for the US series (1971–2024).

Impulse response functions

Orthogonalised impulse response functions (OIRFs) trace the effect of a one-standard-deviation structural shock in each predictor on GDP Growth over a ten-year horizon. Cholesky decomposition is used for identification, with the ordering: Trade/GDP $\rightarrow$ Investment/GDP $\rightarrow$ Inflation $\rightarrow$ GDP Growth (least- to most-endogenous). Bootstrapped 90% confidence bands are drawn from 300 replications.

An investment shock has a persistent positive effect on GDP Growth that peaks around years 2–3 before tapering off. The inflation shock produces an initial negative dip, consistent with contractionary monetary policy responses to rising prices. The trade shock effect is smaller in magnitude and less precisely estimated, echoing the near-zero cross-sectional correlation observed in Section 3.

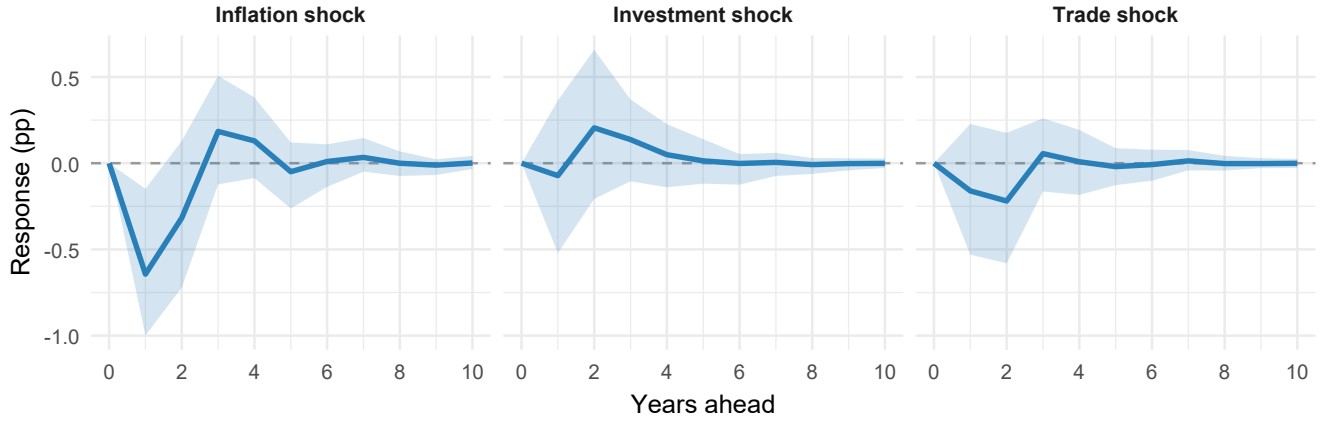


Figure 3: Orthogonalised impulse response functions: effect of a one-SD structural shock on US GDP Growth, with 90% bootstrapped confidence bands (300 replications). Dashed line at zero.

Granger causality

Table 4: Granger causality tests: $H_0 =$ cause does not Granger-cause GDP Growth (US VAR, $p = 2$ lag(s)).

Cause	F-statistic	df	p-value
Investment/GDP	1.205	6, 172	0.306
Inflation (log)	0.496	6, 172	0.811
Trade/GDP	0.844	6, 172	0.537

None of the three predictors Granger-causes GDP Growth at the 5% level under this specification. The Granger framework tests predictive content rather than structural causation; the OIRFs above are needed to characterise the sign and persistence of the identified shocks.

Conclusions

The four analyses in this report converge on a consistent picture. The **panel unit root tests** (Section 2) find that most WDI indicators are stationary in levels over 1990–2024, validating the levels specifications used in Sections 3 and 4. The **two-way fixed-effects regression** (Section 3) confirms conditional convergence, a positive investment-growth link, and a negative inflation effect—all robust to controlling for both country-specific heterogeneity and global time shocks. The **ML ensembles** (Section 3) deliver modest but genuine out-of-sample R^2 improvements over OLS, with large errors concentrated in crisis years, confirming that GDP growth is fundamentally hard to forecast from annual macroeconomic fundamentals alone. The **VAR impulse responses** (Section 4) corroborate the regression findings in a fully dynamic framework: an investment shock generates a persistent multi-year growth dividend, an inflation shock produces an initial contraction, and the trade shock effect is small and imprecisely estimated—consistent with the near-zero correlation identified in the EDA.

Two distributional features dominate all modelling

choices. GDP per Capita and Inflation are heavily right-skewed and require log transformation. Financial-depth variables (Credit/GDP, Market Cap/GDP) are disproportionately missing for lower-income economies, so any model including these regressors implicitly conditions on the existence of formal capital markets—a selection that should be made explicit rather than left buried in the sample.

Researchers extending this dataset in the Dollar & Kraay (2002) tradition should plan for: (i) GMM instrumentation to address dynamic endogeneity in the panel; (ii) quantile regression to test whether the investment-growth link differs across the growth distribution; (iii) country-specific VARs for additional representative economies to assess generalisability of the US VAR findings; (iv) cointegration analysis for variables that prove non-stationary, replacing the VAR in levels with a Vector Error Correction Model (VECM).

References

- Dollar, D., & Kraay, A. (2002). Growth is good for the poor. *Journal of Economic Growth*, 7(3), 195–225.
- Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica*, 46(6), 1251–1271.
- King, R. G., & Levine, R. (1993). Finance and growth: Schumpeter might be right. *Quarterly Journal of Economics*, 108(3), 717–737.
- Levine, R. (2005). Finance and growth: Theory and evidence. In *Handbook of Economic Growth* (Vol. 1A, pp. 865–934). Elsevier.
- Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer.
- Maddala, G. S., & Wu, S. (1999). A comparative study of unit root tests with panel data and a new simple test. *Oxford Bulletin of Economics and Statistics*, 61(S1), 631–652.
- Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1), 1–48.
- World Bank. (2026). *World Development Indicators*

[Dataset]. World Bank DataBank. Retrieved 24 February 2026.

Wright, M. N., & Ziegler, A. (2017). ranger: A fast implementation of random forests for high dimensional data in C++ and R. *Journal of Statistical Software*, 77(1), 1–17.